

OFFICIAL ARCHITECTURE & USER MANUAL

NYAYAAI

RESEARCH. ANALYZE. VERIFY. DRAFT.

Comprehensive technical handbook, advocate user guide, and feature specification for the NYAYAAI Legal Intelligence Platform. Includes full dual-law compliance (BNS / BNSS / BSA vs IPC / CrPC / IEA), cloud deployment procedures, and ethical guidelines.

Version: 1.0.0 Production Release • **Target Jurisdiction:** Republic of India

Date: September 2026 • **Tech Stack:** Next.js 14, TypeScript, Tailwind, Gemini

AI-Powered Indian Legal Intelligence Platform

Tagline: "Research. Analyze. Verify. Draft."

Version: 1.0.0 (Production Release) — September 2026

1. Executive Summary & Vision

NYAYAAI is an enterprise-grade AI legal intelligence platform designed specifically for Indian advocates, senior counsels, law chambers, and legal scholars. The platform tackles the fundamental operational bottlenecks in Indian legal practice:

- The Dual-Law Era:** Seamless transition and comparative analysis between India's historic criminal codes (**IPC 1860**, **CrPC 1973**, **IEA 1872**) and the 2023 enactments (**Bharatiya Nyaya Sanhita [BNS]**, **Bharatiya Nagarik Suraksha Sanhita [BNSS]**, and **Bharatiya Sakshya Adhiniyam [BSA]**).
- Zero-Hallucination Legal Research:** A 5-stage research pipeline producing 9-part structured legal briefs with verified statutory and precedential citations.
- Citation & Precedent Verification:** Verification of Supreme Court and High Court citations (AIR, SCC, SCR) with real-time "Good Law", "Partially Verified", and "Overruled" status tags.
- Judgment Analyzer:** Automated segregation of binding **Ratio Decidendi** from persuasive **Obiter Dicta**, with dual-mode views for Senior Counsels and Junior Advocates.
- Procedural Case Management:** Interactive visual timelines tracking cases from initial filing to final decree.
- Intelligent Legal Drafting Studio:** 14 standardized Indian legal templates with multi-format export (.DOCX, .PDF, .TXT).
- Bar Council of India (BCI) Compliance:** Non-solicitation adherence, strict client confidentiality, and cryptographic SHA-256 audit logging.

2. How to Run the Platform

2.1 Local Developer Environment

Prerequisites:

- **Node.js:** v18.17.0+ or v20.x or v22.x
- **npm:** v9+ or v10+
- **Git:** Installed and configured

Quick Start Commands:

```
# 1. Clone the repository
git clone https://github.com/SUBHAPRIYAM-dev/-NyayaAI-AI-Legal-Research-Case-Intelligence-Platform-for-Indian-Advocates-
cd -NyayaAI-AI-Legal-Research-Case-Intelligence-Platform-for-Indian-Advocates-

# 2. Install dependencies
npm install

# 3. Start local development server
npm run dev
```

Open **http://localhost:3000** in your web browser. The app runs immediately with pre-loaded demo data.

Quality & Verification Commands:

- `npm test` — Executes the complete Vitest test suite (19/19 tests passing).
- `npm run typecheck` — Validates TypeScript types across the entire project (0 errors).
- `npm run build` — Compiles the optimized Next.js production build for all 35 routes.
- `npm run db:seed` — Seeds the in-memory database with 22 statutes, 22 judgments, 22 cases, 10 documents, and 10 research sessions.

2.2 Cloud Deployment on Vercel

1. Log in to vercel.com using your GitHub credentials.
2. Click **"Add New..."** → **"Project"**.
3. Import the repository: `-NyayaAI-AI-Legal-Research-Case-Intelligence-Platform-for-Indian-Advocates-`.
4. **Important:** Change the **Project Name** in Vercel to `nyayaai` (Vercel disallows names starting or ending with hyphens).
5. In **Environment Variables**, add:
 - `JWT_SECRET` : `nyayaai-super-secure-jwt-secret-key-32-chars-min-change-in-prod`

- `COOKIE_NAME` : `nyayaa_session`
- `NODE_ENV` : `production`
- `GEMINI_API_KEY` : (Optional) Add your Google AI Studio key, or leave blank to run in offline Demo AI Mode.

1. Click **Deploy**. Vercel compiles and hosts the app on a live HTTPS URL.

3. How a User Operates NYAYAAI: Step-by-Step User Journey

Step 1: Authentication & Chambers Setup

- **1-Click Demo Login**: Visit `/login` and click either:
- **"Demo Advocate"**: Instantly signs in as Adv. Arjun Sharma (Senior Partner, Supreme Court practice).
- **"Demo Firm Admin"**: Signs in as Chambers Administrator with system health telemetry and audit access.
- **New Advocate Registration**: Register at `/register` and complete the 5-step Onboarding Wizard at `/onboarding` (Bar Council enrollment, chambers structure, practice areas, citation style, and confidentiality settings).

Step 2: Navigating the Executive Dashboard (`/dashboard`)

- Use the **Global Command Palette** (`Cmd+K` or `Ctrl+K`) to jump directly to any matter, statute, or tool.
- View upcoming Supreme Court and High Court hearing dates, recent research queries, active litigation matters, and AI quota meters.

Step 3: 3-Column AI Legal Research Engine (`/research`)

- **Left Column**: Filter past research sessions by practice area (Criminal, Corporate, Constitutional, Taxation).
- **Center Column**: Enter any natural language legal question (e.g., "*Anticipatory bail standards under BNSS Section 482 vs CrPC Section 438*").
- **5-Stage Progress Visualizer**:
 1. *Understanding legal proposition...*
 2. *Identifying jurisdiction & active statutes...*
 3. *Searching statutory provisions (BNS / IPC)...*
 4. *Verifying citations against official reporters...*
 5. *Compiling structured legal brief...*
- **9-Part Structured Legal Brief**:
 1. **Legal Issue**: Clear statement of the legal controversy.
 2. **Short Answer**: Direct affirmative or negative answer under current law.
 3. **Applicable Law**: Dual statutory provisions (BNS vs IPC).
 4. **Relevant Authorities**: Reported Supreme Court and High Court citations.
 5. **In-Depth Analysis**: Structured IRAC (Issue, Rule, Application, Conclusion) legal reasoning.
 6. **Supporting Authorities**: Favorable precedents with bench holdings.

- 7. **Contrary Authorities:** Distinguishable cases and conflicting precedents.
- 8. **Practical Considerations:** Filing tactics, limitation issues, and evidentiary prerequisites.
- 9. **Unresolved Questions:** Gray areas or matters pending before a Larger Bench.
- **Actions:** One-click **Copy Brief**, **Export (.TXT)**, **Create Case Docket**, or **Prepare Draft**.

Step 4: Citation & Precedent Verification (/citation-checker)

- Input a case name or citation (e.g., 2024 INSC 412).
- Inspect verification status:
- **VERIFIED** — Good Law, binding under Article 141 of the Constitution.
- **PARTIALLY VERIFIED** — Distinguished on facts or limited statutory scope.
- **UNVERIFIED** — Overruled or non-existent citation (hallucination safeguard).

Step 5: Judgment Analyzer & Ratio Decidendi Extractor (/judgment-analyzer)

- Upload or select any lengthy judgment.
- Automatically segregates:
- **Ratio Decidendi:** The binding legal principle.
- **Obiter Dicta:** Incidental judicial remarks.
- **Submissions:** Appellant vs Respondent arguments.
- Switch between **Senior Chambers Brief** and **Junior Advocate Walkthrough**.

Step 6: Case Docket & Visual Procedural Timeline (/cases & /cases/[id])

- Manage case metadata: CNR Number, Court Hall, Bench, Stage, and Next Hearing Date.
- Interactive chronological roadmap:
 Filing → Notice Issued → Written Statement → Framing of Issues → Evidence → Final Hearing → Judgment & Decree .
- Log new orders and adjourned hearing dates directly into the case timeline.

Step 7: Case Strategy & Risk Matrix (/case-strategy)

- Evaluate litigation posture across 4 quadrants:
- **Procedural Risk:** Limitation periods, CPC Order VII Rule 11 grounds.
- **Precedent Risk:** Adverse rulings, pending Larger Bench references.
- **Evidentiary Risk:** Electronic evidence certificates (BSA Section 63).
- **Financial Risk:** Costs under CPC §35A, deposit requirements.

Step 8: Document Vault & Contextual AI Reader (/documents)

- Split-screen reader: view contracts, petitions, and orders on the left.
- Contextual AI actions on the right: **Find Legal Risks**, **Find Provisions**, **Find Authorities**, and **Explain Clause**.

Step 9: Legal Drafting Studio (/drafting)

- Choose from **14 standardized Indian legal templates**:
 1. Anticipatory Bail Petition (BNSS §482 / CrPC §438)
 2. Regular Bail Application (BNSS §480 / CrPC §437)
 3. Legal Notice for Cheque Dishonour (NI Act §138)
 4. Commercial Suit Complaint (Commercial Courts Act)
 5. Writ Petition (Constitution Art. 226 / 32)
 6. Special Leave Petition (Constitution Art. 136)
 7. Written Statement / Defence (CPC Order VIII)
 8. Section 9 Interim Measures (Arbitration Act §9)
 9. Caveat Petition (CPC §148A)
 10. Statement of Claim (Arbitration Act §23)
 11. Criminal Revision Application (BNSS §438 / CrPC §397)
 12. Consumer Dispute Complaint (CPA 2019 §35)
 13. Trademark Opposition Notice (Trade Marks Act §21)
 14. Criminal Quashing Petition (BNSS §528 / CrPC §482)
- Select drafting tone: *Aggressive*, *Neutral*, *Conciliatory*, or *Firm*.
- Export in 1-click to **.DOCX**, **.PDF**, and **.TXT**.

Step 10: Firm Workspace & Role-Based Access Control (/firm)

- Manage chambers team members with roles: **Partner** , **Associate** , and **Admin** .
- Invite colleagues and assign matter permissions.

Step 11: Billing & AI Quota Metering (/billing)

- Real-time consumption meters for Queries, Documents, Drafts, Citations, and Tokens.
- Compare tiers: Chambers Starter, Senior Advocate Pro, Enterprise Chambers.

Step 12: Enterprise Administration Console (/admin)

- Real-time subsystem health status (Database, Redis, Gemini AI, Vector Store, Auth).
 - Runtime feature flag toggles.
 - Immutable SHA-256 cryptographic audit logs with tenant filtering.
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4. Complete Feature Inventory (35 Routes)

Module	Route	Type	Core Functionality
Marketing Landing Page	/	Page	Showcase, BNS comparison, pricing, trust badges
Authentication	/login , /register , /forgot-password	Pages	JWT auth, bcrypt passwords, demo shortcuts
Onboarding Wizard	/onboarding	Page	5-step chambers setup wizard
Executive Dashboard	/dashboard	Page	Hearings, dockets, research feed, AI quota meters
AI Legal Research	/research	Page	3-column workspace, 5-stage simulation, IRAC brief
Citation Checker	/citation-checker	Page	Precedent verifier (AIR, SCC), Good Law badges
Judgment Analyzer	/judgment-analyzer	Page	Ratio Decidendi vs Obiter Dicta, dual persona view
Litigation Dockets	/cases , /cases/[id]	Pages	CNR search, procedural timeline, event logger
Case Strategy	/case-strategy	Page	4-factor risk matrix, tactical recommendations
Document Vault	/documents	Page	Split-screen reader, contextual risk & clause analysis
Drafting Studio	/drafting	Page	14 Indian legal templates, tone selector, export
Authorities Library	/authorities	Page	BNS/BNSS/BSA vs IPC/CrPC/IEA comparative browser
Firm Management	/firm	Page	Law chamber team roster, RBAC (Partner/Associate/Admin)
Billing & Quotas	/billing	Page	Quota meters, token tracking, 1-click upgrades
User Settings	/settings	Page	Advocate profile, bar enrollment, preferences
Admin Console	/admin	Page	System health, feature flags, audit trail, indexing
API Backend	17 route handlers in /api/*	API	Serverless REST endpoints with tenant isolation

5. Ethical, Regulatory & Security Framework

- **Bar Council of India Rule 36 Compliance:** Strict non-advertising posture; operations confined to internal chambers research.
- **Zero-Hallucination Policy:** Visual warnings on unverified citations and watermarks on synthetic demonstration records.

- **Mandatory Disclaimers:** All drafts and analyses include statutory notices requiring advocate verification prior to court filing.
- **Cryptographic Audit Trail:** Every user action is logged with an immutable SHA-256 hash for enterprise compliance audits.

NYAYAAI AI-Powered Indian Legal Intelligence Platform • Confidential & Proprietary Manual • September 2026 • Compliant with Bar Council of India Standards